

Alpilla 600 MW Combined Cycle Power Plant

I&C Design Criteria and Control Philosophy

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1. Purpose and scope

1.1 This document defines the instrumentation and control (I&C) design criteria and the control philosophy of the Alpilla 1 x 1 combined cycle plant: the control system architecture and its integration with the turbine control systems (TCS) of the power island supplier, automation level, redundancy rules, the safety instrumented system, signal standards, field instrument and control valve criteria, alarm management, cybersecurity, time synchronisation, emission monitoring, historian and the operator training simulator.

1.2 Scope: the plant DCS (00CJA), remote I/O cabinets, the balance-of-plant safety instrumented system (00CJG), the historian, performance monitoring and simulator (00CKA), the continuous emission monitoring system (00CQA), telecommunication and security systems (00CYA), and all field instruments, analysers and control valves of the plant. The internal control of the gas turbine, steam turbine and generator is by the TCS of the power island supplier (IEC); this document defines only the interface and the rules that apply to the whole plant.

1.3 Out of scope: electrical protection and switchgear control (ALP-EPC-00000-EL-DBR-0001), process design criteria and instrument data sheets; detail design of loops is in the DCS functional design specification (ALP-EPC-00000-IC-SPC-0005), the alarm management philosophy (ALP-EPC-00000-IC-RPT-0001), the OT cybersecurity concept (ALP-EPC-00000-IC-RPT-0002) and the BOP SIS safety requirements specification (ALP-EPC-00000-IC-SPC-0006).

1.4 Status: values marked preliminary are IEPC design criteria that hold until the DCS functional design specification is issued; values marked assumption need confirmation by the DCS supplier, the TCS supplier or a technical query.

2. Governing requirements

ER-12.01: Single integrated plant DCS for GT, ST, HRSG and BOP with redundant controllers, networks and power supplies; operation of the whole plant from the central control room by two operators.

ER-12.02: Automatic start-up and shutdown sequences (one-button start to base load) and automatic load control from the dispatch set-point.

ER-12.03: SIS per IEC 61511 with SIL verification, including burner management and HRSG protection.

ER-12.04: Certified continuous emission monitoring (NO_x, CO, O₂, NH₃, flow, moisture) per EN 14181 with data transmission to the environmental authority.

ER-12.05: Plant historian (5 years on-line storage) and on-line performance calculation per ASME PTC 46 principles.

ER-12.06: High-fidelity operator training simulator of the whole plant, delivered 6 months before first fire.

ER-12.07: OT cybersecurity per IEC 62443 (target security level SL2), zones and conduits, DMZ, patch and backup management, security acceptance test.

ER-12.08: GPS time synchronisation of DCS, protection relays and recorders (1 ms sequence of events).

ER-14.06: Hazardous area classification per IEC 60079-10-1; equipment certified accordingly.

ER-14.07: HAZID, HAZOP, SIL determination, fire and explosion risk assessment and a Major Accident Prevention assessment where applicable.

2.1 Related requirements that the criteria below also address: ER-05.06 (remote diagnostics of the gas turbine with secure OEM access), ER-14.02 (emission limits), ER-14.03 (noise limit of 85 dB(A) at 1 m) and ER-15.03 (factory acceptance tests). Where this document conflicts with a requirement the requirement governs and a technical query is raised.

3. Applicable design data

Parameter	Value	Condition	Status	Basis
Ambient temperature - maximum for full capability	40 degC		confirmed	ER-02.02
Ambient temperature - minimum for full capability	-8 degC		confirmed	ER-02.02
Ambient temperature - survival range	-15 to +45 degC		confirmed	ER-02.02
Annual mean temperature (measured)	14.9 degC	2015-2024 site mast	confirmed	SRC-OWNR-0001
Chloride deposition maximum	420 mg/m2/day	ISO 9225, category S3	confirmed	SRC-OWNR-0001, ER-02.10
Ground flash density	2.4 flashes/km2/year	26 thunderstorm days/year	confirmed	SRC-OWNR-0001, ER-02.12
Performance test ambient range	5 to 30 degC		confirmed	SRC-OWNR-0001, ER-02.11
PM10 maximum (event)	260 ug/m3	Saharan dust episodes	confirmed	SRC-OWNR-0001, ER-02.10
Record maximum / minimum temperature	39.6 / -11.4 degC	1960-2024	confirmed	SRC-OWNR-0001
Sea-salt aerosol annual mean	14 ug/m3	as NaCl	confirmed	SRC-OWNR-0001
Sea-salt aerosol maximum (event)	165 ug/m3	as NaCl, lodos storms	confirmed	SRC-OWNR-0001, ER-02.10
SRC ambient dry-bulb temperature	15 degC	Site Reference Conditions	confirmed	ER-02.03
SRC barometric pressure	1,011.5 mbar	Site Reference Conditions	confirmed	ER-02.03
SRC relative humidity	70 %	Site Reference Conditions	confirmed	ER-02.03
Summer design dry bulb (0.4 %)	32.8 degC	coincident RH 48 %	confirmed	SRC-OWNR-0001, ER-02.11
Winter design dry bulb (99.6 %)	-4.8 degC	coincident RH 88 %	confirmed	SRC-OWNR-0001, ER-02.11
CO	30 mg/Nm3	dry, 15 % O2, 50-100 % GT load	confirmed	ER-14.02
CO on LDO	50 mg/Nm3	dry, 15 % O2, 70-100 % GT load	confirmed	SRC-OWNR-0007, ER-14.02
Dust on LDO	5 mg/Nm3	dry, 15 % O2	confirmed	SRC-OWNR-0007, ER-14.02
NH3 slip	5 mg/Nm3	dry, 15 % O2	confirmed	ER-14.02
NOx (as NO2)	30 mg/Nm3	dry, 15 % O2, 50-100 % GT load	confirmed	ER-14.02
NOx on LDO	50 mg/Nm3	dry, 15 % O2	confirmed	SRC-OWNR-0007, ER-14.02
SO2 on LDO	10 mg/Nm3	dry, 15 % O2	confirmed	SRC-OWNR-0007, ER-14.02
Stack height	65 m	EIA decision (approved dispersion study)	confirmed	ER-06.06, SRC-OWNR-0007
Near-field noise	85 dB(A)	1 m from equipment	confirmed	ER-14.03
Noise at nearest receptor (day)	55 dB(A)	LAeq day	confirmed	SRC-OWNR-0007, ER-14.04
Noise at nearest receptor (night)	45 dB(A)	LAeq night	confirmed	ER-14.04
Cold start time to base load	180 min	maximum	confirmed	ER-04.04
Design life	30 years		confirmed	ER-01.03
Hot start time to base load	45 min	maximum	confirmed	ER-04.04
Operating hours per year	8,000 EOH/year		confirmed	ER-01.03
Starts per year	250 starts/year	50 cold, 100 warm, 100 hot	confirmed	ER-01.03
Warm start time to base load	90 min	maximum	confirmed	ER-04.04
100-year coastal still water level	0.9 m	national datum; +1.40 m with 0.5 m sea-level rise; floor levels >= +1.90 m (ER-02.09); shoreline run-up +4.8 m	confirmed	SRC-OWNR-0002, ER-02.09
Atmospheric corrosivity category	C5 (ISO 12944-2), durability H -		confirmed	ER-02.08
Basic wind speed	30 m/s	10 min mean, 10 m, 50-year return	confirmed	ER-02.05
Design rainfall intensity	110 mm/h	10 min, 25-year return	confirmed	ER-02.05
Distance to heavy-lift port	18 km by sea	400 t quay crane	confirmed	SRC-OWNR-0005
Ground snow load	0.75 kN/m2		confirmed	ER-02.05
Groundwater level	13.5 m a.s.l.	brackish, chloride 3,000-8,000 mg/l	confirmed	SRC-OWNR-0008
Heavy transport road limit	120 t gross	two bridges on the state road	confirmed	SRC-OWNR-0005, ER-02.14
Height datum	TUDKA-99 orthometric heights (m a.s.l.)		confirmed	SRC-EPCE-0011
Plant grid origin in UTM zone 35N (E 0 / N 0)	E 541,250.000 m / N 4,532,480.000 m	TUREF (ITRF96), south-west plot corner	confirmed	SRC-EPCE-0011
Platform level acceptance tolerance	0.15 m	Owner-levelled platform +15.00 m	confirmed	DEC-OWNR-0002
Plot dimensions	400 x 300 m	12.0 ha; usable about 10.6 ha	confirmed	SRC-OWNR-0005, ER-02.01

Parameter	Value	Condition	Status	Basis
Rotation plant grid to UTM grid	-0.3281 deg	plant north = true north; grid convergence at site	confirmed	SRC-EPCE-0011
Scale factor plant grid to UTM	0.999612 -	combined UTM point scale and height factor	confirmed	SRC-EPCE-0011
Seismic design PGA (DD-2, 475-year)	0.45 g	TBDY 2018, site-specific hazard study	confirmed	ER-02.06
Seismic site class	ZD (Vs30 = 240 m/s) -	TBDY 2018	confirmed	SRC-OWNR-0008
Site grade level	15 m a.s.l.		confirmed	ER-02.01

3.1 Field instruments and control equipment shall be rated for the ambient range of DP-CASE-0007 to DP-CASE-0009, the marine C5 atmosphere (DP-CASE-0003) with the salt and chloride loads of DP-OWNR-0005 and DP-OWNR-0007, and the seismic design of DP-CASE-0002. Seawater-wetted parts shall be designed for the chloride content and temperature of DP-OWNR-0014 and DP-CASE-0013 and the intermittent chlorination of DEC-EPCE-0015.

3.2 Start-up times to base load (DP-CASE-0037 to DP-CASE-0039), the load ramp rate (DP-CASE-0032) and the minimum environmental load (DP-CASE-0033) are the performance targets of the automatic sequences and load control of section 6; the sequences shall be tuned and demonstrated against them.

4. Codes and standards

4.1 Functional safety: IEC 61511 (process sector) with IEC 61508 for device certification. Cybersecurity: IEC 62443 (all parts). Hazardous areas: IEC 60079-10-1, ATEX 2014/34/EU. Emission monitoring: EN 14181, EN 15267-3, EN 15259. Combustion and boiler protection: NFPA 85 (functions performed in the TCS and the SIS). Alarm management: EEMUA 191 and IEC 62682 / ISA-18.2. Control valves: IEC 60534. Enclosures: IEC 60529. Communication: IEC 61850, OPC UA (IEC 62541).

4.2 Performance calculations per ASME PTC 46 principles, with ASME PTC 22 (GT), PTC 6 (ST), PTC 4.4 (HRSG) and PTC 12.2 (condenser) as the basis of the on-line calculation methods.

4.3 Tagging and classification of instruments, loops and control functions shall follow KKS and the AVEVA class library of the Owner (ER-01.06); the instrument index (ALP-EPC-00000-IC-LST-0001) is the single source of the instrument data. Standards named in this document but not in the project register are proposed for registration (see findings).

5. Control system architecture and philosophy

5.1 One integrated plant DCS shall control and monitor the balance of plant (BOP): fuel gas and LDO systems, HRSG auxiliaries, water-steam cycle auxiliaries, cooling water, water treatment, compressed air, HVAC, electrical systems, fire detection and utilities. The GT, ST and HRSG main protection and control functions are executed by the TCS of the power island supplier, integrated in the same operator environment (ER-12.01, TQ-026).

5.2 Integration with the TCS: all TCS operation, alarms, trends and events are presented on the DCS operator stations with the DCS look and feel and with no separate operator HMI in normal operation; the TCS remains hardware of the power island supplier and keeps the turbine protection. The interface is via redundant OPC UA between the TCS servers and the DCS, with one KKS tag database exported to the DCS, plus hardwired signals for start, stop, emergency trips and the load set-point (TQ-026 and TQ-IEC-004).

5.3 The TCS shall be a separate IEC 62443 zone (section 13), connected to the DCS through a conduit with firewall; the OPC UA connection uses signed and encrypted sessions with certificates. A failure of the OPC UA link shall not affect the turbine control or protection; the DCS shall show the loss of link on the operator screens and revert to the hardwired signals for start / stop / trip.

5.4 Hardwired signals between DCS and TCS are limited to the ones needed for safe and autonomous operation of each system: start / stop commands, permissives, emergency stop (redundant), ready signals, run signals and trip-outs; the unit load set-point and the export limiter signal are transmitted in redundant analogue signals and as OPC UA values (ALP-EPC-00000-IC-CNR-0001).

5.5 The plant DCS shall not be used to execute functions of a certified SIL rating; those are in the SIS (section 8) or in the TCS protection system of the supplier. The DCS shall provide the plant-wide sequences, closed loops and monitoring of the BOP and the coordination of the units (unit coordination, load control, export limiter, fuel transfer).

5.6 Network architecture: a redundant plant network (ring or dual star with rapid recovery, 100 ms or less, to the controllers); separate networks for control (level 1), supervisory (level 2), the remote I/O, the safety network, and an OT DMZ (level 3.5) towards the Owner business network and remote access; each network shall be physically separate switches or VLAN with firewalls between zones.

5.7 Control room: one central control room (CCR) from which the whole plant is operated by two operators (ER-12.01). The CCR shall have four operator workstations (two per operator, each with three screens), one shift supervisor workstation, one engineering workstation with access to the DCS and TCS engineering, one large screen display of the overview, and printers only for the alarm and event logs on demand. Field operator interface is by local panels and mobile devices with restricted rights.

5.8 Local control panels shall exist only for packages delivered with own PLC (e.g. SWRO, condensate polishing, chlorination, gas station, LDO unloading, compressors, water treatment); each package PLC shall be operated from the DCS through a serial or Ethernet link (Modbus TCP or OPC UA) and shall run its sequences autonomously; the local panel shall be the exception, used for maintenance and commissioning.

5.9 All emergency shutdown push buttons in the CCR (unit trip, fuel gas and LDO ESD, fire) shall be hardwired to the SIS or TCS independent of the DCS network. The DCS shall never be the only means to trip the plant.

6. Automation level and operating modes

6.1 Automatic start-up and shutdown sequences shall take the unit from standstill to base load with a single operator command (one-button start) for the hot, warm and cold start of DP-CASE-0037 to DP-CASE-0039, with the sequence steps (GT start, HRSG warm-up, steam admission, ST start, load ramp, emission compliance, SCR ammonia injection) coordinated by the unit coordination logic; the operator can hold, step or abort a sequence at each defined step.

6.2 Automatic load control from the dispatch set-point (or the operator set-point) with the ramp rate of DP-CASE-0032 and the limitations of the minimum environmental load (DP-CASE-0033), the export limiter of the connection capacity (DP-OWNR-0051) and the primary and secondary frequency control of the Grid Code; the ramp and limiter logic is in ALP-EPC-00000-IC-CNR-0001.

6.3 Automation levels: Level 1 individual drive and valve control (open / close, start / stop with interlocks); Level 2 group control (subsystem sequences with permissives, start of a redundant pump group on standby); Level 3 unit coordination (start-up, load control, shutdown, fuel transfer). Every level can be operated from the operator station in manual, automatic or step mode; a failure of a higher level shall leave the lower levels in the last state and operable.

6.4 Standby and duty-changeover of redundant equipment (2 x 100 %, 2 x 50 % + 1) shall be automatic: standby start on low discharge pressure or flow or on trip of the running unit within the process dead time, with alarm; equal running hours shall be balanced by selectable lead / lag.

6.5 Trip philosophy: process trips shall bring the plant to the safe state without an operator action; the hierarchy of trips is unit trip (GT, ST), HRSG trip, system trip and drive trip. Trip first-out and sequence of events at 1 ms resolution are recorded; a trip reset shall need a confirmed operator action after the cause has cleared.

6.6 Manual operation of every automatic function shall be possible from the DCS; process-critical equipment (fire water, lube oil, turning gear, emergency drives) has a hardwired manual start / stop in the CCR or local panel that overrides the DCS in an emergency and is monitored by the DCS.

6.7 Operator interface: DCS graphics based on the KKS structure, uniform colour and symbol standard (grey background, colour reserved for abnormal condition), navigation depth of three levels at most, overview, unit, system and detail displays; trends with 1 s resolution for tags on trip conditions; response of display calls of 2 s or less for 95 % of the calls and update of dynamic values of 1 s.

7. Redundancy and availability rules

Item	Rule	Remarks
DCS controllers	Redundant (hot standby), bumpless transfer, one pair per process unit or function group	A single failure shall not cause a loss of control or a trip [ER-12.01]
DCS controller loading	Controller CPU load 60 % or less and memory 60 % or less at commissioning	Reserve for the growth and the peak load of the scan [ER-12.01]
Plant and I/O networks	Redundant, with automatic failover in 100 ms or less; no single switch or cable failure shall isolate a controller or station	[ER-12.01]
I/O modules	Redundant I/O modules and communication for the loops that trip or shut down a system; non-redundant modules with segregation of redundant equipment for the others	Signals of 2 x 100 % duty equipment on different I/O modules and different remote I/O cabinets [ER-12.01]
Power supplies	Redundant power supplies (2 x 100 %) fed from two independent UPS sources (00BRA10) with monitoring; 24 V DC by redundant modules with diode / redundancy decoupling	Each cabinet is fed by two feeders from the two UPS distributions [ER-12.01, ER-11.05]
Operator stations and servers	Any operator station can perform any function of an operator; server and historian redundant	[ER-12.01]

Item	Rule	Remarks
Spare capacity	20 % spare I/O channels and 20 % spare slots and cabinet space; 20 % spare controller capacity	Equal to the I/O count basis of the DCS equipment record [ER-12.01]
Voting for trips	2oo3 for trip sensors of the protective functions; 1oo2 where the SIL calculation allows	Separate impulse lines and process taps for each sensor; the measurement for control is a fourth sensor or the median of the three [ER-12.03]
Redundant sensors for control	Median selection of 3 sensors or the average of 2 with deviation alarm for the loops whose failure would trip the unit	Signal validation and deviation alarm (2 % of span, 5 s) [ER-12.01]

7.1 Failure of any single component of the DCS (controller, I/O module, power supply, network, workstation, server) shall cause no trip, no loss of view and no loss of control of any process. A failure shall be alarmed and displayed on the system diagnostics display; a power loss of one UPS branch shall not affect the DCS. The DCS supplier shall demonstrate this in the factory acceptance test (ER-15.03) by failure injection.

7.2 Final elements shall go to the fail-safe position defined in the P&ID on loss of signal, loss of instrument air or loss of power; outputs of a failed controller go to the configured fail position; motors are not restarted automatically after a restore of the DCS (except the process-critical drives with an automatic restart defined in the functional design specification).

8. Safety instrumented systems

8.1 The safety instrumented system (SIS) shall be designed per IEC 61511 with SIL verification, including burner management (NFPA 85) and HRSG protection functions (ER-12.03). The SIS shall be physically and logically separate from the DCS: own logic solver, own I/O, own networks, own power supply, own engineering station, own sensors and final elements for each safety instrumented function.

8.2 The BOP SIS logic solver (00CJG10) shall be a redundant, IEC 61508 certified logic solver (2oo3 architecture, SIL 3 capable) for the BOP functions: fuel gas and LDO emergency shutdown, ammonia, chlorine / hypochlorite, fire and gas detection and other functions from the SIL determination. The GT, ST and HRSG protection functions supplied with the TCS are executed by the certified protection system of the power island supplier; the allocation of every function to the TCS or to the BOP SIS is fixed in the safety requirements specification (ALP-EPC-00000-IC-SPC-0006).

8.3 The SIL of every safety instrumented function is determined by the HAZOP and layer of protection analysis (ER-14.07, ALP-EPC-00000-HS-HAZ-0007) and verified by calculation (PFDavg and architecture constraints, proof test interval, diagnostic coverage, common cause factor). The safety requirements specification defines function, safe state, SIL, response time, demand mode, proof test interval, bypass and reset rules.

8.4 Design rules for the SIS: de-energise-to-trip; sensors are independent from the control sensors (no shared transmitter); voting by 2oo3 for the safety functions of SIL 2 or higher and for the functions whose spurious trip means a unit trip; separate impulse lines; final elements dedicated to the SIS (no shared valve with control) with partial stroke testing and diagnostics, where possible; the safety response time budget shall be at most 50 % of the process safety time.

8.5 Devices in a SIF shall have an IEC 61508 certificate for the SIL, or prior-use justification per IEC 61511; the safety manual of each device applies; the installation shall not degrade the SIL.

8.6 The SIS shall be connected to the DCS by a redundant read-only data link (Modbus TCP or OPC UA) for indication and alarms; no command from the DCS to the SIS except reset and maintenance override through the dedicated, key-locked and alarmed route; the SIS engineering station is connected only to the SIS network.

8.7 Bypass and override: every SIS bypass is by a keylocked switch, alarmed on the DCS, time-limited by an administrative procedure and logged in the sequence of events; no permanent bypass; the automatic reset of a trip is not allowed. Proof tests are done on the intervals of the SIL verification and recorded.

8.8 Fire and gas detection shall be in the SIS with the mapping of the fire and gas detector study (ALP-EPC-00000-HS-STU-0003): voting 2oo3 for the executive actions, low alarm at 10 %, high alarm at 20 % and executive action at 40 % of the lower explosive limit for methane and hydrogen (as HS-DBR-0001 section 10); ammonia, chlorine and other toxic gases per the HAZOP results; detectors are certified for the zone (ER-14.06) and for the marine atmosphere.

8.9 Sequence of events: all SIS and TCS trips are time-stamped in the I/O modules at 1 ms resolution and delivered to the DCS sequence-of-events recorder; the SOE shall be stored for the required period of the historian (section 16).

9. Signal standards and communication

9.1 Analogue field signals shall be 4-20 mA, two-wire, loop-powered at 24 V DC, with HART (version 7). The failure signals per NAMUR NE43 (below 3.6 mA and above 21 mA are faults). HART variables and diagnostics are transferred to the asset management system of the DCS; device status per NAMUR NE107.

9.2 Field instruments and control valve positioners use 4-20 mA HART; PROFIBUS PA and FOUNDATION Fieldbus are not used for field devices and PROFIBUS DP is not used between the DCS and the packages (DEC-EPCE-0017). Reasons: (a) the diagnostics of the instruments are available by HART without fieldbus segments; (b) a wiring failure affects a single loop, not a segment of several instruments and valves; (c) fewer protocols reduce the engineering, spare parts, testing and the cybersecurity attack surface of the IEC 62443 conduits; (d) the TCS integration is by OPC UA. Package PLCs and smart devices communicate by Modbus TCP or OPC UA; MV protection relays by IEC 61850.

9.3 Digital signals: dry contacts wetted at 24 V DC (or interposing relays for 220 V DC or 230 V AC signals) with line monitoring for the safety and critical signals; open circuit and short circuit monitoring for signals of trips and emergency stops.

9.4 Start, stop, trip and status of every motor and valve are hardwired to the MCC or the switchgear interface; diagnostics and measurements of smart MCC and drives are by Modbus TCP over redundant networks and are not used for control or interlock.

9.5 Temperature: Pt100 resistance thermometers class A (IEC 60751) in 3-wire or 4-wire connection, thermocouples type K (or N above 1,000 degC) to IEC 60584; transmitter mounted in the head or in the field for temperatures used in control loops. Analogue signals shall be wired in individually screened twisted pairs, screens earthed at one end (the DCS side).

9.6 Hazardous areas: intrinsically safe (Ex i) circuits with isolators / barriers for zone 0 / 1 instrument loops, or Ex d instruments; all field devices in zones are certified per ER-14.06. Ex i loops are documented with entity parameters and cable capacitance checks.

9.7 Surge protection at the ends of the signal cables leaving buildings or laid outdoors over long distances (lightning protection zones of ALP-EPC-00000-EL-STU-0004); DCS cabinets conform to IEC 61000-6-2 and IEC 61000-6-4 electromagnetic compatibility.

10. Field instrument criteria

Item	Criterion	Remarks
Pressure and differential pressure	Smart transmitters, accuracy 0.075 % of span or better, static-pressure effect and stability per datasheet; overpressure rating not less than 1.5 x the design pressure of the process	Impulse lines with three-valve manifold; remote seals only where the process needs them [ER-12.01]
Temperature	Pt100 class A in thermowells sized for the wake frequency (ASME PTC 19.3 TW), immersion length according to the process; thermocouples for above 400 degC	Accuracy of the loop of 0.5 % of span or better [ER-12.01]
Level	Guided wave radar or DP with wet leg for the vessels; HRSG drum level by three DP transmitters (2oo3) plus independent measurement for the SIS	Sensors on separate nozzles [ER-12.03]
Flow	Orifice / venturi (ISO 5167) with DP transmitter for steam, feedwater and gas; electromagnetic for conductive water; ultrasonic or vortex for others; fiscal gas metering as in the fuel gas station	Meter uncertainty for performance test flows per ASME PTC 46 principles [ER-12.05]
Range selection	Normal operating value between 50 % and 70 % of the calibrated range; alarm and trip set points between 10 % and 90 %	[ER-12.01]
Enclosure and ingress protection	IP66 / IP67 outdoors; junction boxes IP66; DCS cabinets in air-conditioned rooms IP42 minimum	IEC 60529 [REF-CONS-0092]
Exposed housing and tubing materials	Stainless steel 316L minimum; super-duplex or 6Mo (PREN 40 or above) for tubing, fittings and enclosures within the coastal C5 zone and near seawater and chlorination systems; aluminium housings only with C5 coating	ISO 12944-2 C5-H for coated parts [REF-CONS-0092, DP-CASE-0003, DP-OWNR-0007]
Seawater and chlorinated water wetted parts	Titanium grade 2, super-duplex (UNS S32750) or 6Mo; not 316L	Chloride per DP-OWNR-0014; residual oxidant per DEC-EPCE-0015 [DP-OWNR-0014, DP-CASE-0013, DEC-EPCE-0015]
Ambient temperature	-15 degC to +45 degC ambient, +70 degC sun-heated surface; heat tracing or heated enclosures for impulse lines below 0 degC	Freeze protection with insulation and self-limiting heating cable, monitored [DP-CASE-0009, DP-CASE-0007]
Safety-related devices	IEC 61508 certified for the SIL, or prior use per IEC 61511	[REF-CONS-0081]

10.1 All instruments have a KKS tag, a datasheet generated from the instrument index and a calibration certificate; smart devices are configured by HART DTM / EDD and are documented in the asset management system. Local indicators shall be provided for every local measurement needed by the field operator, not as a substitute for a transmitter.

10.2 Analysers: steam and water analysis system (00QUA10) per VGB-S-010 and DEC-EPCE-0015: cation conductivity, specific conductivity, pH, dissolved oxygen, sodium and silica at the sampling points of the cycle chemistry; two independent cation conductivity measurements per condenser water box half (ER-07.10); sample coolers and constant temperature and pressure conditioning; alarms and limits of the chemistry are a design input from the cycle chemistry criteria.

10.3 Vibration, position and speed monitoring of BOP machines (pumps, fans, compressors above 200 kW) with proximity probes or accelerometers evaluated per ISO 20816-3 where the machine size warrants; the machinery protection of the GT and ST is in the TCS.

10.4 Safety valves and instruments for pressure protection that are not SIS functions are independent of the DCS; instruments for custody transfer and performance testing shall be of test quality with calibration traceable to national standards and the uncertainty in the performance test procedure (ER-12.05).

11. Control valve sizing criteria

11.1 Control valves shall be sized per IEC 60534-2-1 (liquids, gases and steam) using the fluid properties and flows of all HMB cases and the process design criteria; the maximum case governs the flow (DEC-EPCE-0014), the normal and minimum cases shall be shown. The rated flow coefficient shall be the value for the maximum flow with a margin of 10 %.

11.2 Valve position and rangeability: at maximum flow the valve shall be open 85 % or less of the travel (80 % for butterfly), at minimum flow 10 % or more; the installed characteristic shall keep the gain of the loop within a factor of 2 over the range; the inherent rangeability of the valve of 50:1 or more for globe valves, 30:1 for rotary valves, and split range or parallel valves where the required rangeability is higher.

11.3 Pressure drop: the control valve differential pressure at normal flow shall be 25 % or more of the dynamic pressure drop of the system and at least 0.7 bar for liquid service; valves for pump minimum flow, bypass and spray are sized on their governing case (all-cases envelope).

11.4 Noise: the sound pressure level shall not exceed 85 dB(A) at 1 m from the valve and the pipe (ER-14.03, DP-CASE-0044) in normal operation, calculated per IEC 60534-8-3 (aerodynamic) and IEC 60534-8-4 (hydrodynamic) without credit for insulation unless specified as part of the supply; low-noise trim, multistage trim, diffusers or a series of valves where the calculated level is higher; the manufacturer shall guarantee the level and measure it in the performance test when required.

11.5 Cavitation and flashing: continuous cavitation above the incipient-damage limit is not permitted; anti-cavitation trim or several stages where the pressure drop exceeds the limit; hardened trim for flashing and erosive services; outlet velocity of gas at 0.3 Mach or lower (0.5 Mach at the maximum flow for short durations), liquid outlet velocity of 8 m/s or lower, steam according to the design criteria of the piping.

11.6 Leakage: seat leakage class IV per IEC 60534-4 for control valves, class V or VI (bubble-tight) for tight shutoff, isolation and SIS valves; SIS shutdown valves are tight shutoff and fire safe, with partial stroke testing and a fail-safe spring return in the safe direction. Fugitive emission from stems for fuel gas services shall be tested per ISO 15848-1 class B or better.

11.7 Actuators and positioners: pneumatic spring-return actuators with fail position on loss of air and signal as on the P&ID; sized for the maximum differential pressure of the closed valve with a margin of 25 % at the minimum air supply pressure; digital positioners with HART and diagnostics; full stroke time 10 s or less for control valves (3 s or less for the turbine bypass and spray-water attemperation valves, subject to the TCS supplier), stroke time and fail position stated in the datasheet.

11.8 Materials and construction: body materials per the piping class of the service; seawater services with titanium, superduplex or rubber / ceramic-lined bodies; exterior coating per ISO 12944-2 C5-H; stainless steel 316L tubing and fittings, 6Mo where in the coastal zone; instrument air filter regulators of stainless steel.

11.9 Control valve datasheets and sizing shall be generated from the valve and instrument records; the manufacturer shall provide the sizing, noise, cavitation and actuator calculations for every valve with the tender, and shall state the flow at the three cases (minimum, normal, maximum).

12. Alarm management

12.1 Alarm management follows EEMUA 191 and IEC 62682 / ISA-18.2, as accepted in TQ-026 for the TCS alarms: one plant alarm philosophy for the DCS, TCS and package alarms (ALP-EPC-00000-IC-RPT-0001); an alarm is configured only if it requires an operator action, has a documented cause, consequence, corrective action and response time (rationalisation), and has a priority.

Item	Target	Remarks
Priority distribution	High 5 %, medium 15 %, low 80 % of the configured alarms	Three priorities; a new alarm needs justification of the priority [ER-12.01]
Average alarm rate per operator (steady state)	1 alarm per 10 min or less (manageable); never more than 2 per 10 min	Measured over 30 days of stable operation [ER-12.01]
Alarm flood	10 alarms in 10 min or less after an upset; a flood (more than 10 alarms in 10 min) shall not exceed 1 % of the time	[ER-12.01]
Standing alarms	10 stale (standing) alarms or fewer at any time; no more than 5 alarms displayed as high priority at once	[ER-12.01]
Chattering alarms	None; deadband, on-delay and off-delay set per alarm; repeating alarms shall be corrected within the maintenance period	[ER-12.01]

12.2 Design measures: state-based alarming (alarms suppressed automatically for stopped equipment and during start-up and shutdown sequences, with the suppression rule documented), grouped alarms and first-out for trips, alarm shelving with a time limit and authorisation, no alarm on a signal that is already an alarm of the parent equipment, and alarm text of the standard structure (tag, condition, action).

12.3 Alarm system performance (KPIs of EEMUA 191) shall be measured and reported monthly from the historian, and the alarm list (ALP-EPC-00000-IC-LST-0002) is under change control; a change of limit, priority or delay follows the management of change with the operator approval. The KPIs shall be met at the operational demonstration (ER-16.06) and reported.

12.4 Critical alarms of the SIS and the fire and gas system are annunciated by hardwired alarm annunciators or separate displays in the CCR in addition to the DCS, with distinct audible tones for the priorities.

13. Cybersecurity

13.1 OT cybersecurity shall be designed per IEC 62443 with the target security level SL-T 2 for every zone, as required by ER-12.07: zones and conduits per IEC 62443-3-2, security requirements of the systems per IEC 62443-3-3 (SL-C at or above SL-T), secure product development and component requirements of IEC 62443-4-1 and 4-2 demonstrated by the suppliers, and security acceptance test.

13.2 Zones: DCS (level 1 / 2), TCS (own zone, TQ-IEC-004), SIS (own zone, no direct connection to the business network), package PLCs, CEMS and DAHS, historian and simulator, engineering and remote access, telecom and security systems (CCTV, access control), and an OT DMZ (level 3.5) between the plant network and the Owner network. Conduits have industrial firewalls with default-deny rules, the allowed protocols and directions documented.

13.3 Remote access (e.g. OEM diagnostic access of ER-05.06) only through the OT DMZ with a jump host, multi-factor authentication, time-limited approval by the operator, session recording, and no direct connection from the internet to a control network; no wireless network in the control zones unless the security assessment permits it.

13.4 System hardening: role-based access with individual accounts (no shared accounts), password policy, removal of unused services and ports, USB and removable media control, application whitelisting on the servers and workstations, and malware protection compatible with the DCS supplier; every component has a documented baseline configuration.

13.5 Patch and backup management: security patches are assessed by the suppliers within 30 days of the release and installed after a test on the offline test system; backups of all controller and server configurations after every change and at least weekly on the servers, with an offline copy; restore tested at least annually; logs sent to a central log server in the DMZ.

13.6 Security acceptance test per ALP-EPC-00000-IC-TST-0002 with the report ALP-EPC-00000-IC-TRP-0001: verification of zone separation, firewall rules, hardening, access control, backup and restore, with vulnerability scanning of the systems; findings above the accepted risk are closed before the plant handover.

14. Time synchronisation

14.1 GPS time synchronisation of the DCS, TCS, protection relays, disturbance recorders, SIS and CEMS with sequence of events at 1 ms resolution (ER-12.08). Two redundant GPS master clocks (GNSS receivers with holdover) are installed in the CCR building; the GPS antennas are lightning protected and shielded against interference.

14.2 Distribution: PTP (IEEE 1588, power profile IEEE C37.238) for the protection relays and TCS (as offered in TQ-IEC-004: TCS synchronised from the DCS master clock by PTP), NTP for servers and workstations, IRIG-B for recorders; accuracy of 1 ms or better at every time stamping device, and holdover accuracy of 1 ms or better for 24 hours after the loss of GPS.

14.3 Time base: UTC in all systems and storage, with the local time (UTC+3, no daylight saving) at the display; a loss of synchronisation is alarmed. Time-stamping of digital inputs for the SOE is in the input module.

15. Continuous emission monitoring (CEMS)

15.1 The CEMS on the stack (00CQA10) measures NOx, CO, O₂, NH₃, flow and moisture (ER-12.04), and dust when firing LDO, with stack temperature and pressure for the conversion of the values to the reference conditions (dry, 15 % O₂, standard conditions) of the emission limits of DP-CASE-0040 to DP-CASE-0042 (gas) and DP-OWNR-0041 to DP-OWNR-0043 (LDO).

15.2 Automated measuring systems (AMS) shall be certified per EN 15267 (QAL1) for the measured components, with a certified range not larger than 1.5 times the emission limit where a certified range is available; the measuring range in use shall let the exceedances be shown and the fuel changeover (gas to LDO, with different limits) be handled in the DAHS.

15.3 Quality assurance per EN 14181: QAL2 (calibration and variability test by an accredited laboratory after installation), QAL3 (control charts with automatic zero and span check), AST (annual surveillance test by an accredited laboratory); the measurement location and sampling per EN 15259 with permanent access platform, sampling ports and a homogeneous sample point at the stack (10HNE10 with CEMS platform).

15.4 Redundancy: dual NOx and O₂ analysers with automatic change-over so that an analyser failure does not lead to loss of valid data; availability of valid hourly means of 95 % or more; heated sample line and conditioning (extractive systems) with an alarm on the temperature and flow; calibration gases in certified cylinders with the flow control by the DAHS.

15.5 Data acquisition and handling system (DAHS): calculation of the validated hourly, daily and annual means per the LCP Regulation and the environmental permit with the uncertainty deduction, storage of at least 5 years, reports for the authority, and transmission of the data to the environmental authority by the online link required by the permit; the DAHS is in its own zone (section 13) and signals to the DCS by hardwired 4-20 mA for the operator display and the control of the SCR.

15.6 The NOx and NH₃ signals from the CEMS are used for the SCR ammonia dosing supervision only; the SCR control uses its own analysers or the TCS model, so that a CEMS calibration does not disturb the process control.

16. Historian, performance monitoring and operator training simulator

16.1 Plant historian: a redundant historian server with 5 years of on-line storage (ER-12.05); all DCS, TCS, SIS, package and CEMS signals are archived, including all TCS signals (TQ-026), with the capacity for the number of archived tags of the historian record (00CKA10) plus 20 % spare; time-stamped at the source; data are not compressed for trip-relevant tags (a 1 s or faster recording, and high-resolution 100 ms snapshot on a trip).

16.2 Historian sampling and compression: analogue tags at 1 s scan with an exception deadband of 0.1 % of span, compression deviation of 0.2 % of span; digital tags on change; calculated performance tags every 60 s; the archives after 5 years go to offline storage for the design life period per the Owner's data policy; access to the Owner is read-only through the OT DMZ.

16.3 On-line performance monitoring per ASME PTC 46 principles: the plant net output, net heat rate, GT, ST, HRSG, condenser and cooling water performance, corrected to the reference conditions and compared with the guarantee curves of the HMB and the vendor data; the calculations shall use validated measurements (deviation alarm and signal validation) and the calculation methods of the test codes (section 4); results shall be displayed to the operator and stored in the historian.

16.4 Operator training simulator (OTS): a high-fidelity, whole-plant simulator (ER-12.06) that contains the same DCS graphics and control functions (emulated controllers or the hardware of the DCS) and the TCS logic (or a validated model),

with dynamic models of the GT, HRSG, ST, condenser, cycle, cooling water, water treatment, fuel systems and the electrical system; instructor station with malfunction, snapshot, replay and scenario functions; the specification is ALP-EPC-00CKA-GE-SPC-0001.

16.5 Fidelity and delivery: steady-state agreement with the HMB cases of 2 % or better on the main process variables and correct dynamic behaviour (trends and time constants) within 5 %; a minimum of 200 malfunctions; delivered 6 months before first fire (ER-12.06) and updated to the final logic before the start of the commissioning; the simulator is used to test the start-up sequences, the trip logic and the operator procedures before the site tests.

17. Power supply, installation and testing of I&C equipment

17.1 I&C cabinets shall be installed in air-conditioned rooms (N+1 cooling, positive pressure, filtered air per ER-10.06), separated by function (DCS, SIS, TCS, CEMS) with the SIS in separate cabinets; cable entry from below with the fire and gas sealing; cabinet heat dissipation and temperature rise are part of the room HVAC design.

17.2 Cabling: instrument, signal and communication cables per the cable criteria of the electrical design criteria (halogen-free flame-retardant inside buildings, fire-rated for the safety functions, separate trays from power cables, spare cores 20 %); fibre-optic cable for the networks between buildings, duplicated on separate routes.

17.3 Factory acceptance test of the DCS and the SIS with the complete configuration (graphics, logic, sequences, alarms, redundancy failure injection, cybersecurity baseline) per ER-15.03; site acceptance per ALP-EPC-00000-IC-TST-0001 with the loop checks of 100 % of the I/O and the functional tests of every sequence and trip; SIL proof of the SIS installation and the validation per IEC 61511.

18. Equipment covered by this document

Tag	Description	Service	Redundancy	Rating	Design P / T	Material
00CJA10	Plant DCS (controllers, networks, operator stations)	Plant DCS: controllers, redundant networks, operator/engineering stations	Redundant controllers and networks	10,600 I/O points (incl. 20 % spare), 230 V		Steel cabinets
00CJA20	DCS remote I/O cabinets - power block	DCS remote I/O cabinets - power block	1 x 100 % (redundant I/O power supply)	4,600 I/O points (incl. 20 % spare), 230 V		Steel cabinets IP54, C5 coating
00CJA30	DCS remote I/O cabinets - CW pump house	DCS remote I/O cabinets - CW pump house	1 x 100 % (redundant I/O power supply)	800 I/O points (incl. 20 % spare), 230 V		Steel cabinets IP54, C5 coating
00CJA40	DCS remote I/O cabinets - water treatment	DCS remote I/O cabinets - water treatment	1 x 100 % (redundant I/O power supply)	1,800 I/O points (incl. 20 % spare), 230 V		Steel cabinets IP54, C5 coating
00CJA50	DCS remote I/O cabinets - BOP utilities	DCS remote I/O cabinets - BOP utilities	1 x 100 % (redundant I/O power supply)	2,000 I/O points (incl. 20 % spare), 230 V		Steel cabinets IP54, C5 coating
00CJA60	DCS remote I/O cabinets - fuel gas and LDO	DCS remote I/O cabinets - fuel gas and LDO	1 x 100 % (redundant I/O power supply)	1,400 I/O points (incl. 20 % spare), 230 V		Steel cabinets IP54, C5 coating
00CJG10	BOP safety instrumented system (logic solver and cabinets)	BOP safety instrumented system logic solver and cabinets (SIL 2, IEC 61511)	Redundant logic solver (2oo3)	300 I/O points (incl. 20 % spare), 230 V		Steel cabinets
00CKA10	Historian, performance monitoring and operator training simulator	Historian, performance monitoring and operator training simulator	Redundant servers (historian)	25,000 archived tags, 230 V		Steel racks
00CQA10	Continuous emission monitoring system	Continuous emission monitoring system (HRSG stack: NOx, CO, O ₂ , NH ₃ slip, flow, dust)	1 x 100 % (dual NOx/O ₂ analysers)	7 analysers, 230 V		Heated sampling line PTFE / 316L, analyser shelter
00CYA10	Telecom, CCTV, access control and security system	Telecom, CCTV, access control and perimeter security	1 x 100 %	60 CCTV cameras (plus access control, PABX, radio), 230 V		Steel racks / weatherproof housings
00QUA10	Steam/water analysis system (SWAS)	Steam/water sampling and analysis (SWAS) panel	1 x 100 %	16 sample points, 20 kW, 400 V	10 barg / 50 degC	Stainless steel 316L tubing and panels
00QUA10AC001	Sample cooler rack	Sample cooler rack (high-pressure sample coils, CCW cooled)	1 x 100 %	20 kW	250 barg / 620 degC	Stainless steel 316L coils

18.1 The equipment above is the I&C equipment of the plant as recorded in the equipment records at the time of issue; the rated data are those of the records (single source), not of this document.

References

Codes and standards cited in this document (Codes, Standards and Regulations Register ALP-EPC-00000-GE-LST-0001; latest edition at contract date unless stated).

Code	Title
Env. Permit Regulation	Environmental Permit and Licence Regulation (Cevre Izin ve Lisans Yonetmeligi)
LCP Regulation	Large Combustion Plants Regulation (Buyuk Yakma Tesisleri Yonetmeligi)
ATEX 2014/34/EU	ATEX Equipment Directive, transposed as the regulation on equipment for potentially explosive atmospheres
ASME PTC 22	Performance Test Code on Gas Turbines
ASME PTC 6	Performance Test Code on Steam Turbines
ASME PTC 4.4	Performance Test Code on Gas Turbine Heat Recovery Steam Generators
ASME PTC 12.2	Performance Test Code on Steam Surface Condensers
NFPA 85	Boiler and Combustion Systems Hazards Code
ASME PTC 46	Performance Test Code on Overall Plant Performance
VGB-S-010	Feed water, boiler water and steam quality for power plants
NFPA 72	National Fire Alarm and Signaling Code
IEC 61850	Communication networks and systems for power utility automation
IEC 61511	Functional safety - Safety instrumented systems for the process industry sector
IEC 62443	Security for industrial automation and control systems (all parts)
IEC 60079-10-1	Explosive atmospheres - Classification of areas - Explosive gas atmospheres
EN 14181	Stationary source emissions - Quality assurance of automated measuring systems
EN 15267-3	Air quality - Certification of automated measuring systems - Part 3
EN 15259	Air quality - Measurement of stationary source emissions - Measurement sections and sites
ISO 12944-2	Paints and varnishes - Corrosion protection of steel structures by protective paint systems - Part 2: Classification of environments